

Topology

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Abstract

These are the definitions and theorems from Ben Fitzpatrick's topology sequence MH 650, MH 651, MH 652 for the Auburn University academic year 1970-1971 as transcribed by his student John W. Bales. Much editorial assistance was provided by fellow student Sam Creswell who made his notes available as well.

The following material is from Professor Ben Fitzpatrick's topology sequence taught at Auburn University in the 1970-71 academic year. The original numbering of the theorems has been maintained. Where Dr. Fitzpatrick inserted additional theorems into the sequence we have retained the numbering which he gave to such extra theorems, if any. Although the definitions were not originally numbered, we have numbered them in the order in which they were given in class. With regard to Theorem Z, it was Dr. Fitzpatrick's custom to give it a designation outside the regular numbering sequence.

"Point" and "region" are undefined.

Axiom (0). *Every region is a point set.*

Definition 1. *The statement that the point P is a limit point of the set M means that if R is a region containing P as an element, then R contains a point of M different from P .*

Definition 2. *The statement that the point P is a boundary point of the point set M means that if R is a region containing P , then R contains a point of M and a point not in M .*

Notation. *If M is a point set, then $\overline{M}$ denotes the point set to which P belongs if and only if P is a point of M or a limit point of M .*

Definition 3. *The statement that the point set M is closed means that if P is a limit point of M , then P belongs to M .*

Theorem 1. *No point of a region is a boundary point of that region.*

Theorem 2. *If M is a point set, then $\overline{M}$ is closed.*

Definition 4. *An order relation on a set M is a collection Ω of ordered pairs of elements of M with the following properties:*

1. *If x and y are two elements of M , then (x, y) or (y, x) is an element of Ω .*
2. *If (x, y) and (y, z) are elements of Ω , then (x, z) is an element of Ω .*
3. *If x is in M , then (x, x) is not in Ω .*

If Ω is an order relation on M , then the element x of M is said to precede the element y of M according to Ω if and only if (x, y) is in Ω .

Definition 5. A well-ordering on M is an order relation Ω on M such that if H is a subset of M , then there is an “first” element of H , i.e. an element x of H such that if y is any other element of H , then x precedes y according to Ω .

Definition 6. The collection G of point sets is said to be monotonic if and only if it is true that if g and h are two members of G , then either g is a subset of h or h is a subset of g .

Definition 7. The pointset M is perfectly compact if and only if it is true that if $\mathcal{H}$ is a monotonic collection of subsets of M , then there is a point P which is either common to all the members of $\mathcal{H}$ or is a limit point of every member of $\mathcal{H}$.

Theorem 3. If G is a collection of regions covering the closed and perfectly compact point set M and Ω is a well-ordering on G , then there exists a member y of G such that the collection of regions in G which y does not precede according to Ω covers M .

Theorem ($\mathbb{Z}$). If G is a non-degenerate set, then there exists a well-ordering on G .

Theorem 4. If G is a collection of regions covering the closed and perfectly compact point set M , then some finite sub-collection of G covers M .

Axiom (1_3). There exists a sequence $G_1, G_2, G_3, \dots$ such that

1. if n is a positive integer, then G_n is a collection, each member of which is a region, and G_n covers S .
2. If n is a positive integer, G_{n+1} is a sub-collection of G_n .
3. if R is a region, A is a point of R and B is a point of R , then there exists a positive integer n such that if g is a region of the collection G_n and g contains A , then $\bar{g}$ is a subset of R and does not contain B unless B is A .

Theorem 5. If P is a point, then there exists a sequence $R_1, R_2, R_3, \dots$ such that

- (i) if n is a positive integer, then R_n is a region containing P
- (ii) if n is a positive integer, then $\bar{R}_{n+1}$ is a subset of R_n .
- (iii) P is the only point common to all the terms of the sequence $R_1, R_2, R_3, \dots$
- (iv) if R is a region containing P , then there is a positive integer n such that R_n is a subset of R .

Definition 8. The sequence $R_1, R_2, R_3, \dots$ in the previous theorem is said to “close down” on the point P .

Theorem 6. If P is a point and G is a finite collection of regions each containing P , then there is a region containing P which is a subset of each member of G .

Theorem 7. *If a point P is a limit point of the sum of two point sets H and K , then it is a limit point of one of them.*

Theorem 8. *No finite point set has a limit point.*

Theorem 9. *If the point P is a limit point of the point set M , then there exists a sequence $P_1, P_2, P_3, \dots$ of points of M all distinct from P and from one another such that P is the sequential limit point of the sequence $P_1, P_2, P_3, \dots$.*

Definition 9. *The statement that the point P is the sequential limit point of the point sequence $P_1, P_2, P_3, \dots$ means that if R is a region containing P , then there is a positive integer N such that if k is a positive integer greater than or equal to N , then P_k is a point of R .*

Definition 10. *The statement that the point set M is bi-compact means that if G is a collection of regions covering M , then a finite sub-collection of G covers M .*

Theorem 10. *Every bi-compact point set is closed and perfectly compact.*

Theorem 11. *No point sequence has two sequential limit points and if the point P is the sequential limit point of the point sequence $P_1, P_2, P_3, \dots$, then P is the sequential limit point of every infinite subsequence of $P_1, P_2, P_3, \dots$.*

Theorem 12. *If α is a point sequence and the set of all terms of α is compact, then some subsequence of α has a sequential limit point.*

Definition 11. *The statement that the point set M is compact means that if K is an infinite subset of M , then K has a limit point.*

Theorem 13. *If the set of all terms of the infinite point sequence α is compact and there do not exist two points each of which is a sequential limit point of some subsequence of α , then α has a sequential limit point.*

Definition 12. *The statement that the point set D is a domain means that if P is a point of D , then there is a region which contains P and is a subset of D .*

Theorem 14. *If H and K are two mutually exclusive closed point sets and one of them is perfectly compact, then there exist mutually exclusive domains D_H and D_K containing H and K respectively.*

Theorem 15. *If M is a point set and G is a collection of regions covering $\overline{M}$, then there exists a subset K of M with no limit point such that*

- (1) *no region of G contains two points of K and*
- (2) *if P is a point of M , then some region of G contains P and intersects K .*

Theorem 16. *If G is a collection of domains covering the bi-compact point set M , then some finite sub-collection of G covers M .*

Definition 13. A point set M is said to be separable if and only if there is a countable subset K of M such that every point of M is a point of K or a limit point of K . The point set M is said to be completely separable if and only if there is a countable collection G such that each element of G is a domain and such that if P is a point of M and R is a region containing P , then some domain of the collection G contains P and is a subset of R .

Theorem 17. If M is a point set, then the following statements are equivalent:

1. M is completely separable.
2. Every subset of M is separable.
3. If K is an uncountable subset of M , then K has a limit points.
4. If G is a collection of domains covering M , then some countable sub-collection of G covers M .

Theorem 18. If H and K are two mutually exclusive closed point sets and G is a monotonic collection of closed sets and each element of G intersects both H and K and no element of G is the sum of two mutually exclusive closed point sets, one intersecting H and the other intersecting K , and some element of G is perfectly compact, then the common part of all the elements of G is closed and perfectly compact and intersects both H and K and is not the sum of two mutually exclusive closed point sets, one intersecting H and the other intersecting K .

Definition 14. The statement that the point sets H and K are mutually separated means that they do not intersect and neither of them contains a limit point of the other.

Definition 15. The point set M is connected if and only if M is not the sum of two mutually separated point sets.

Theorem 19. If the connected point-set M is the subset of the sum of two mutually separated point sets, then it is a subset of one of them.

Notation. If G is a collection of sets, then G^* denotes the set to which P belongs if and only if P is an element of some set in G .

Theorem 20. If G is a collection of connected point sets and some element of G intersects every other element of G , then G^* is connected.

Definition 16. If each of M and N is a point set, then the statement that T is a transformation from M onto N means that T is a collection each element of which is an ordered pair whose first term is an element of M and whose second term is an element of N and no two elements of T have the same first term and every element of M is the first term of some member of T and every element of N is the second term of some member of T .

The statement that the transformation from M onto N is continuous means that if x is a point of M and ϵ is a region containing $T(x)$, then there exists a

region δ containing x such that if q belongs to the common part of δ and M , then $T(q)$ is in ϵ .

Theorem 21. *If T is a continuous transformation from M onto N , then*

- (1) N is connected if M is,
- (2) N is bi-compact if M is, and
- (3) N is separable if M is.

Theorem 22. *Every perfectly compact point set is compact and every compact point set is perfectly compact.*

Theorem 23. *Every point of a non-degenerate connected point set is a limit point of that point set.*

Theorem 24. *If M is connected and M is a subset of L and L is a subset of $\overline{M}$, then L is connected.*

Theorem 25. *If H and K are two point sets and P is a boundary point of their common part, then P is a boundary point of one on them.*

Definition 17. *If M is a point set, then the statement that C is a component of M means that C is a connected subset of M and is not a proper subset of any connected subset of M .*

Theorem 26. *If M is a point set, then every point of M belongs to a component of M .*

Theorem 27. *If M is a point set and n is a positive integer such that M has only n components, then M is the sum of n mutually separated point sets*

Definition 18. *A continuum is a closed and connected point set.*

Theorem 28. *If G is a monotonic collection of perfectly compact continua, then the common part of the elements of G is a perfectly compact continuum.*

Definition 19. *Suppose H and K are two mutually exclusive closed point sets and M is a continuum which intersects both H and K , then M is said to be irreducible from H to K .*

Theorem 29. *If M is a closed and perfectly compact point set that intersects both of the two mutually exclusive closed point sets H and K and M is not the sum of two mutually exclusive closed point sets one intersecting H and the other intersecting K but every closed proper subset of M which intersects both H and K is the sum of two such point sets, then M is an irreducible continuum from H to K .*

Theorem 30. *If the perfectly compact continuum M intersects both of two mutually exclusive closed point sets H and K , then some sub-continuum of M is irreducible from H to K .*

Definition 20. A topological transformation is a continuous and reversible transformation T such that T^{-1} is also continuous.

Definition 21. The point set M is topologically equivalent to the point set N if and only if there is a topological transformation from M onto N .

Theorem 31. Topological equivalence is an equivalence relation, i.e.

1. A point set is topologically equivalent to itself.
2. If the point set M is topologically equivalent to the point set N , then N is topologically equivalent to M .
3. If the point set M is topologically equivalent to the point set N and N is topologically equivalent to the point set O , then M is topologically equivalent to O .

Theorem 32. Suppose H and K are two closed point sets and f is a continuous transformation from H to a subset of the point set M and g is a continuous transformation from K to a subset of the point set M . Furthermore, suppose that if x belongs to H and to K , then $f(x) = g(x)$. Then $f \cup g$ is a continuous transformation whose initial set is $H \cup K$.

Theorem 33. If f is a continuous and reversible transformation whose initial set is bi-compact, then f^{-1} is also continuous.

Theorem 34. Suppose f is a transformation with initial set M and x is a point of M , then f is continuous at x if and only if it is true that if $P_1, P_2, P_3 \dots$ is a point sequence which has x as its sequential limit point and each term of this sequence is in M , then the point sequence $f(P_1), f(P_2), f(P_3), \dots$ has $f(x)$ as its sequential limit point.

Theorem 35. If M is a connected point set and T is a connected subset of M and $M - T$ is the sum of two mutually separated point sets H and K , then $H + T$ is connected.

Definition 22. The statement that the point or subset P of the connected point set M separates the point or subset H of M from the point or subset K of M means that $M - P$ is the sum of two mutually separated point sets, one containing H and the other containing K .

Note: In theorems 36, 37 and 38, M denotes a connected point set and P , A , B and C represent points of M .

Theorem 36. If P separates A from B in M , then P separates B from A in M and A does not separate P from B in M .

Theorem 37. If P separates A from C in M and B separates A from C in M , then either P separates A from $\{B, C\}$ in M or B separates A from $\{P, C\}$ in M .

Theorem 38. *If P separates A from B in M and B separates P from C in M , then B separates $\{A, P\}$ from C in M and P separates A from $\{B, C\}$ in M .*

[Note: In Dr. Fitzpatrick's 1970-1971 topology courses, he did not state a theorem 39 or 40.]

Axiom (1). *Consists of Axiom 1₃ plus*

4. *if $M_1, M_2, M_3, \dots$ is a simply infinite sequence, each term of which is a closed point set, and each term of it contains the next term of it, and for each positive number N , M_N is a subset of the closure of some element of G_N , then there is a point common to all the terms of the sequence $M_1, M_2, M_3, \dots$.*

Theorem 41. *If M is a closed point set and every point of M is a limit point of N , then M is uncountable.*

Theorem 42. *Every metric space satisfies Axiom 1₃.*

Theorem (42'). *Every complete metric space satisfies Axiom 1.*

Theorem 43. *Every locally compact 1₃ space satisfies Axiom 1.*

Definition 23. *The statement that a space is locally compact means that if P is a point of the space, then there is a compact region containing P .*

Theorem 44. *No domain is the sum of countable many closed point sets, no one of which contains a domain.*

Theorem 45. *No closed point set M is the sum of countably many closed point sets such that if T is any one of them every point of T is a limit point of M_T .*

Definition 24. *Suppose M is a set and T is a set and Ω_1 is a well-ordering on M and Ω_2 is a well-ordering on T . The statement that Ω_1 is ordinally similar to Ω_2 means that there is a reversible transformation ϕ from M onto T such that if a and b are two elements of M , then a precedes b according to Ω_1 if and only if $\phi(a)$ precedes $\phi(b)$ according to Ω_2 . (i.e. there is a reversible order-preserving transformation from M onto T .)*

Definition 25. *The statement that the subset M' of M is an initial (final) segment of M according to Ω means that there is an element P of M such that M' is the set of all elements of M preceding P (not preceding P) according to Ω .*

Theorem (unnumbered). *If Ω_1 is a well-ordering on M and Ω_2 is a well-ordering on T , then either (1) there is a reversible order-preserving transformation from M onto T or (2) there is a reversible order-preserving transformation from M onto an initial segment of T or (3) there is a reversible order-preserving transformation from an initial segment of M onto T .*

Definition 26. The statement that A is cardinally equivalent to B means that there is a reversible transformation from A onto B . The statement that A is cardinally less than B means that A is cardinally equivalent to a subset of B but is not cardinally equivalent to B . The statement that A is cardinally greater than B means that B is cardinally less than A .

Definition 27. If M is a connected point set and P is a point of M , then P is said to be a separating point of M if $M - P$ is not connected and is a non-separating point of M if $M - P$ is connected.

Definition 28. An arc is a non-degenerate compact continuum with no more than two non-separating points.

Definition 29. The statement that M is an arc from A to B means that M is a compact continuum and if P is a point of M other than A or B , then P is a separating point of M .

Question. If M is an arc are there two points A and B such that M is an arc from A to B ?

Question. Is there a non-degenerate compact continuum M such that no point of M is a non-separating point of M ? That is, is there a non-degenerate compact continuum M such that every point of M is a separating point of M ?

Proposition (A1). If M is an arc then M has two non-separating points.

Proposition (A2). If M is an arc from A to B and P is a point of M different from A and from B , then P separates A from B in M .

Proposition (A3). If M is an arc from A to B , P is a point of M different from A and from B and H and K are two mutually separated point sets whose sum is $M - P$, then A is in H and B is in K or A is in K and B is in H .

Proposition (A4). Under the hypothesis, $H + P$ is an arc from P to A or to B .

Proposition (A5). If M is an arc from A to B , then M is irreducible from A to B .

Proposition (A6). If M is an arc from A to B and $P_1, P_2, P_3, \dots$ is an infinite sequence of points of M and for each positive integer N , P_{N+1} separates the two-point set $\{A, P_N\}$ from B in M , then $P_1, P_2, P_3, \dots$ has a sequential limit point which either is B or separates B from $\{A, P_1, P_2, P_3, \dots\}$ in M .

Theorem (30'). If M is a compact continuum and K is a closed subset of M , then there is a sub-continuum of M which contains K and is such that no proper sub-continuum of it contains K .

Theorem 46. If M is a closed and perfectly compact point set and H and K are two mutually exclusive closed subsets of M and M contains no continuum which intersects both H and K , then M is the sum of two mutually exclusive closed point sets, one containing H and the other containing K .

Theorem 47. *If H and K are two mutually exclusive and closed point sets and M is a compact continuum which is irreducible from H to K , then $M - M \cdot H$ is connected and $M - M \cdot (H + K)$ is connected and every point of $M \cdot H$ is a limit point of $M - M \cdot H$.*

Theorem (unnumbered). *If H is a collection of regions covering S and M is a point set, Z is a subset of M and G is a collection of regions and for each point P of M there exists a region of the collection G that contains every region of the collection H that contains P and if every subset of M that is cardinally greater than Z has a limit point, then some subset of G cardinally less than or equivalent to Z covers M .*

Theorem 48. *If M is a closed and compact point set and K is a component of M and H is a closed subset of $M - K$, then M is the sum of two mutually separated point sets one containing K and the other containing H .*

Theorem 49. *If M is a continuum and D is a domain with respect to M and D is a proper subset of M and $\overline{D}$ is compact and K is a component of D , then the boundary of D with respect to M contains a limit point of K .*

Definition 30. *If M is a point set and D is a subset of M , then the statement that D is a domain with respect to M means that no point of D is a limit point of $M - D$.*

Definition 31. *If T is a subset of the point set M , then the statement that the point X of M is a boundary point of T with respect to M means that every region containing X contains a point of T and a point of $M - T$.*

Proposition (A7). *If T is a continuous and reversible transformation whose initial set M is an arc from A to B , then T is a topological transformation whose final set N is an arc from $T(A)$ to $T(B)$.*

Proposition (A8). *If M is an arc from A to B and P is a point of M different from A and from B and $M - P$ is the sum of two mutually separated point sets H and K and H' and K' are two mutually separated point sets whose sum is $M - P$, then either H' is H and K' is K or H' is K and K' is H .*

Proposition (A9). *Suppose M is an arc from A to B ; $\{P_1, P_2, P_3, \dots\}$ is a countable dense subset of $M - \{A, B\}$; $n_1 = 1$; for each positive integer j , n_{j+1} is the smallest integer $m > n_j$ such that one component of $M - P_m$ contains $\{P_1, P_2, P_3, \dots, P_{n_j}\}$; and P is a limit point of $\{P_1, P_2, P_3, \dots\}$. Then P is a non-separating point of M .*

Definition 32. *The statement that the point set M is the limiting set of the sequence of point sets $M_1, M_2, M_3, \dots$ means that*

1. *there exist a point X such that if R is a region containing X and i is a positive integer, then there is a positive integer j greater than i such that R intersects M_j and*

2. M is the set of all such points.

Theorem 50. If $M_1, M_2, M_3, \dots$ is a sequence of connected point sets and $\overline{M_1 + M_2 + M_3 + \dots}$ is compact and for each positive integer N , a_N is a point of M_N and the point sequence $a_1, a_2, a_3, \dots$ has a sequential limit point, then the limiting set of $M_1, M_2, M_3, \dots$ is a compact continuum.

Theorem 51. If K is a closed proper subset of the continuum M , $\overline{M - K}$ is compact and H is a component of $M - K$, then K contains a limit point of H .

Theorem 52. If K is a component of a closed and compact point set M and T is a closed point set containing no point of K , then

1. there is a domain D containing K such that no point of $T + M$ is a boundary point of D and D contains no point of T and

2. K is a component of $M + T$.

Definition 33. Suppose $M_1, M_2, M_3, \dots$ is a simply infinite sequence each term of which is a point set. Then the statement that the point set M is the sequential limiting set of the sequence $M_1, M_2, M_3, \dots$ means that M is the limiting set of every infinite sub-sequence of $M_1, M_2, M_3, \dots$.

Theorem 53. If the sequence of point sets $M_1, M_2, M_3, \dots$ has a completely separable limiting set, then some infinite sub-sequence of that sequence has a sequential limiting set.

Theorem 54. Every non-degenerate compact continuum contains two non-separating points.

Theorem 55. If M is a compact continuum and H is a connected proper subset of M , then $M - H$ contains a non-separating point of M .

Theorem 56. If M is an arc from A to B , then $M - A$ and $M - B$ are connected.

Theorem 57. If M is an arc from A to B and P is a point of M different from A and from B , then $M - P$ is the sum of two mutually separated point sets and if H and K are two mutually separated point sets whose sum is $M - P$, then one of H and K contains A and the other contains B .

Theorem 58. If M is an arc from A to B and P is a point of M different from A and from B and $M - P$ is the sum of two mutually separated point sets H and K and A is in H , then $H + P$ is an arc from A to P and $K + P$ is an arc from B to P .

Definition 34. The statement that P precedes Q in the order from A to B means that either P is A and Q is B or P separates A from $\{Q, B\}$.

Definition 35. The statement that T is a segment from the point P of the arc M to the point Q of the arc M means that T is the set of points to which the point C belongs only in case C is in M and C precedes Q in M and P precedes C in M (both in the order from A to B in M or both in the order from B to A in M provided M is an arc from A to B .)

Definition 36. If M is an arc from A to B the statement that T is a sect containing A means that there is a point P of M different from A such that T is the subset of M to which the point Q belongs only in case Q precedes P in the order of A to B in M .

Theorem (58.5). If M is an arc and T is a subset of M , then P is a limit point of T only in case either

1. P is a separating point of M and every segment of M containing P contains a point of T different from P or
2. P is a non-separating point of M and every sect of M containing P contains a point of T different from P .

Lemma. If M is an arc from A to B and C is a countable dense subset of M which contains neither A nor B and Z is a countable dense subset of the number segment $(0,1)$, then there is a reversible transformation T from C onto Z which preserves order.

Theorem 59. Every arc is topologically equivalent to the number interval $[0,1]$.

Definition 37. A simple closed curve is a non-degenerate compact continuum each two point subset of which separates it.

Theorem 60. No point of a simple closed curve is a separating point of it.

Theorem 61. If A and B are points of the simple closed curve J , then J is the sum of two arcs from A to B which have in common only A and B .

Theorem 62. If M is a connected point set, then there is no domain D such that the common part of D and M is separable and contains a point set H which contains an uncountable point set K such that if X is any point of K , then $M - X$ is the sum of two mutually separated point sets, one of which contains $H - X$.

Definition 38. The statement that the point set M is connected im kleinen at the point P of M means that if U is an open set containing P , then there is an open set V containing P such that if qQ is a point of the common part of V and M , then some connected set of the common part of U and M contains both P and Q .

Definition 39. The statement that the point set M is open means that M is a domain.

Theorem 63. *If M is connected im kleinen and D is a domain with respect to M , then every component of D is a domain with respect to M .*

Definition 40. *The statement that the collection H of point sets is coherent means that if H_1 and H_2 are two sub-collections of H whose sum is H , then some member of H_1 intersects some member of H_2 .*

Theorem 64. *If $G_1, G_2, G_3, \dots$ is an Axiom 1 sequence for S and for each positive integer n , H_n is a finite coherent sub-collection of G_n and for each n the closure of each element of H_{n+1} is contained in some element of H_n , then the common part of $H_1^*, H_2^*, H_3^*, \dots$ is a compact continuum.*

Theorem 65. *If S is connected and connected im kleinen and A and B are two points of S , then S contains an arc from A to B .*

Theorem (64'). *If M is a connected point set and G is a domain collection covering M and A and B are two points of M , then there is a finite coherent sub-collection of G such that G^* contains both A and B .*

Theorem 66. *If the continuum M is locally compact at the point P of M but is not connected im kleinen at P , then there is a region R containing P and a sequence $K_1, K_2, K_3, \dots$ of distinct components of $M \cdot \overline{R}$ such that each K_n contains a point of R and a point on the boundary of R and $K_1, K_2, K_3, \dots$ converges to a continuum K which contains P and intersects the boundary of R .*